Chang Liu

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Research Interests: Large Language Models, AI Agents, Trustworthy AI

Education

The Hong Kong Polytechnic University **MSc in Blockchain Technology** **Aug 2025 – Present**

- **GPA**: 3.52 (First Semester)
- **Core Courses**: Machine Learning, Distributed Algorithms, Big Data Computing, Electronic Payments.

Dalian Maritime University **BSc in Computer Science and Technology** **Aug 2020 – Jun 2024**

- **GPA**: 83.3 / 100
- **Core Courses**: Operating Systems, Computer Networks, Database Systems, Computer Architecture.

Publications

Academic Papers (ArXiv / Under Review):

- **[1] Me-Agent: A Personalized Mobile Agent with Two-Level User Habit Learning** (first Author/ACL2026)
- **[2] MobileRAG: Enhancing Mobile Agent with Retrieval-Augmented Generation** (first Author/Under Review)
- **[3] DSPrompt: Dynamic Soft Prompt Defense Against M-RAG Corruption** (first Author/AAAI2027 Under Review)

Research Experience

Defense-MMRAG: Defensive Multimodal Retrieval-Augmented Generation Framework **Dec 2025 – May 2026**

Research Core: Multimodal LLM Security via Adversarial Defense Prompts

- Freeze CLIP parameters while inserting learnable **defense prompts** into the document encoder's Transformer layers, employing **PGD-based adversarial training** to precisely filter malicious content without requiring labeled attack data.
- **Result:** Reduces PoisonedEye attack success rate from **89% to 0.5%**, maintains strong generalization across multiple attack types with negligible impact on generation accuracy.

Me-Agent: Personalized Mobile Agent with User Habit Learning **Jun 2025 – Jan 2026**

Research Core: Training-Free GRPO and Hierarchical Memory

- Implemented a Training-Free **GRPO** (Group Relative Policy Optimization) module to learn user preferences by comparing multiple execution trajectories and distilling natural-language experiences.
- Combined with a **HPM** (Hierarchical Preference Memory) that stores app-specific workflows and content preferences for efficient, context-aware retrieval.
- **Result:** Achieved SOTA on User FingerTip benchmark, raising application selection accuracy from 60% to 100% and attaining a general task success rate of **89.3%**.

MobileRAG: Retrieval-Augmented Mobile Agent **Jan 2025 – Jun 2025**

Research Core: Agentic Framework Design and Retrieval-Augmented Generation

- Developed MobileRAG, a comprehensive framework for **LLM-powered mobile agents**, incorporating three novel modules: **LocalRAG** for local application matching, **InterRAG** for external knowledge retrieval, and **MemRAG** for historical experience reuse.
- Achieved a **10.3%** improvement over state-of-the-art baselines on mobile task benchmarks.

Professional Experience

Qiyuan Laboratory **Algorithm Intern (Fundamental AI Department)** **Mar 2024 – Jun 2025**

- **LLM Distributed Training**: Contributed to setting up PyTorch-based distributed training environments for Large Language Models. Responsible for optimizing Supervised Fine-Tuning (SFT) strategies.
- **Performance Optimization**: Executed refined data preprocessing pipelines and deep log analysis, improving the average model accuracy by 8.2% across three major LLM benchmarks.
- **Agent Architecture Research**: Led the internal development and empirical testing of a Mobile Agent system, successfully enabling automated, multi-scenario reasoning and interactions.

Tencent **Algorithm Intern (Tencent Jarvis Lab)** **Jun 2026 – Present**

- Explored vision-language joint reasoning for medical diagnosis via literature review, focusing on visual region grounding and interleaved reasoning chains.
- Designed a multi-agent clinical decision support system with confidence discrimination to enhance medical QA accuracy and enable disease progression monitoring.

Selected Awards & Competitions

- **National Third Prize**, National Computer Design Competition — 2023
  - *Intelligent Traffic Supervision System*: Integrated DeepSORT and HyperLPR for real-time traffic regulation and violation detection.
- **Provincial First Prize**, National Computer Design Competition — 2023
  - *Intelligent Vehicle Assistance Under Severe Weather*: Combined YOLOv5 with Retinex on Jetson Nano/STM32, boosting detection accuracy from 62% to 89%.
- **Outstanding Student Scholarship**, Dalian Maritime University — 2023

Technical & Language Skills

- **Research Focus**: RAG, Autonomous AI Agents, Large Language/Vision Models, Object Detection.
- **Programming**: Python, C/C++, Java, SQL, Solidity, Shell.
- **Frameworks & Tools**: PyTorch, TensorFlow, HuggingFace, Docker, Git, Linux.
- **Hardware & Systems**: Embedded Systems (Jetson Nano, STM32), Web3.0 / Hyperledger Fabric.